I.

$(a, m) = (b, m) = 1$ then $(ab, m) = 1$

$\because$ $(a, m) = 1$ means $ax_1 + my_1 = 1$ $(\exists x_1, y_1 \in \mathbb{Z})$

likewise $bx_2 + my_2 = 1$ $(\exists x_2, y_2 \in \mathbb{Z})$

$1 = (ax_1 + my_1)(bx_2 + my_2)$

$= ab(x_1x_2) + m(ax_1y_2 + bx_2y_1 + my_1y_2)$

therefore 1 is greatest common divisor

2021-10585

서하준

II.

**2.** Find the greatest common divisor $g$ of the numbers 1819 and 3587, and then find integers $x$ and $y$ to satisfy

$$1819x + 3587y = g.$$

$3587 = 1819 \cdot 1 + 1768$

$1819 = 1768 \cdot 1 + 51$

$1768 = 51 \cdot 34 + 34$

$51 = 34 \cdot 1 + 17$

$34 = 17 \cdot 2 + 0$

$1768 = 3587 - 1819 \cdot 1$

$51 = 1819 - 1768 \cdot 1$

$34 = 1768 - 51 \cdot 34$

$17 = -3587 \cdot 36 + 1819 \cdot 71$

$17 = 1819 \cdot 35 - 1768 \cdot 36$

$17 = -1768 \cdot 1 + 51 \cdot 35$

$17 = 51 \cdot 1 - 34 \cdot 1$

$g = 17$

$3587(-36) + 1819(71) = 17$

---

**13.** Prove that $n^2 - n$ is divisible by 2 for every integer $n$; that $n^3 - n$ is divisible by 6; that $n^5 - n$ is divisible by 30.

- $\forall n \in \mathbb{N} \quad 2 \mid n \text{ or } 2 \mid n-1$
- $\forall n \in \mathbb{N} \quad 3 \mid n \text{ or } 3 \mid n-1 \text{ or } 3 \mid n+1$
- $\forall n \in \mathbb{N} \quad 5 \mid n \text{ or } 5 \mid n+1 \text{ or } 5 \mid n-1 \text{ or } 5 \mid n+2 \text{ or } 5 \mid n-2$

- $2 \mid n^2 - n = n(n-1)$ $\quad \forall n \quad 2 \mid n^2 - n$
- $6 \mid n^3 - n = (n-1)(n)(n+1)$ $\quad \forall n \quad 2 \mid n^3 - n \quad 3 \mid n^3 - n$
- $30 \mid n^5 - n = n(n+1)(n-1)(n^2 - 4 + 5)$

  $= 5\,n(n+1)(n-1) + n(n+1)(n-1)(n+2)(n-2)$

  $\forall n \quad 5 \mid n^5 - n$

  $3 \mid n^5 - n$

  $2 \mid n^5 - n$

Since 2, 3, 5 coprime this logic holds

**16.** Prove that if $a$ and $b$ are positive integers satisfying $(a, b) = [a, b]$ then $a = b$.

if $x \mid y$ with $x, y \in \mathbb{Z}$, $x, y > 0$ then $x \le y$

$(a,b) \le \min(a,b) \longrightarrow (a,b) \mid a$ and $(a,b) \mid b$

$\le \max(a,b) \le [a,b] \longrightarrow a \mid [a,b]$ and $b \mid [a,b]$

therefore $\min(a,b) = \max(a,b)$

$= a = b$

---

**24.** Prove that no integers $x, y$ exist satisfying $x + y = 100$ and $(x, y) = 3$.

- $x = 3x_0$, $y = 3y_0$ $(\exists x_0, y_0 \in \mathbb{Z})$
- $x + y = 3(x_0 + y_0) = 100$
- $3 \mid 3(x_0 + y_0)$ but $3 \nmid 100$ $\Rightarrow$ 모순.

---

**43.** Prove that $a \mid bc$ if and only if $\dfrac{a}{(a,b)} \mid c$.

let $a = a'd$, $b = b'd$, $d = (a,b)$

$\therefore$ $(\Rightarrow)$ $a'd \mid b'dc$ then $a' \mid b'c$ since $(a', b') = 1$

by Euclid Lemma $a' \mid c$

$(\Leftarrow)$ $a' \mid c$ then $a'd \mid dc$

then $a'd \mid b'dc$

then $a \mid bc$

***49.** Prove that if $m > n$ then $a^{2^n} + 1$ is a divisor of $a^{2^m} - 1$. Show that if $a, m, n$ are positive with $m \neq n$, then

$$(a^{2^m} + 1, a^{2^n} + 1) = \begin{cases} 1 \text{ if } a \text{ is even} \\ 2 \text{ if } a \text{ is odd.} \end{cases}$$

$$a^{2^m} - 1 = \left(a^{2^{m-1}}\right)^2 - 1$$

$$= (a^{2^{m-1}} + 1)(a^{2^{m-1}} - 1)$$

$$= (a^{2^{m-1}} + 1)(a^{2^{m-2}} + 1) \cdots\cdots (a+1)(a-1)$$

$a^{2^n} + 1 \mid a^{2^m} - 1 \quad (\forall n \in \mathbb{Z} \text{ s.t } 0 \leq n < m)$

---

- gcd is 1 or 2 ∵ $a^{2^m} + 1 = k \cdot (a^{2^n} + 1) + 2$

- 모든 $n \in \mathbb{N}$ 에 대하여 $a^{2^n} + 1$ 은

  홀수일경우는 $a+1$ 은 홀수이기 때문에

Therefore $(a^{2^n} + 1, a^{2^m} + 1) = \begin{cases} 1 & \text{if } a \text{ odd} \\ 2 & \text{if } a \text{ even} \end{cases}$

***51.** Show that if $(a, b) = 1$ and $p$ is an odd prime, then

$$\left(a + b, \frac{a^p + b^p}{a + b}\right) = 1 \text{ or } p.$$

- $a^p = ((a+b) - b)^p$

  $= (a+b)^p - \binom{p}{1}(a+b)^{p-1} b + \cdots + \binom{p}{p-1}(a+b) b^{p-1} - b^p$

- $a^p + b^p = (a+b)^p - \binom{p}{1}(a+b)^{p-1} b + \cdots + p(a+b) b^{p-1}$

- 양변을 $(a+b)$로 나누면 RHS은

  나머지는 $p(a+b)b^{p-1}$가 된다.

i.e $\quad d = \left(a+b, \frac{a^p+b^p}{a+b}\right) = (a+b, pb^{p-1})$

---

- $d = \left(a+b, \frac{a^p+b^p}{a+b}\right)$

- $\frac{a^p+b^p}{a+b} = a^{p-1} - a^{p-2} \cdot b + \cdots + b^{p-1}$

- $b \equiv -a \pmod{d}$

- $\frac{a^p+b^p}{a+b} \equiv p a^{p-1} \pmod{d}$

---

- $k = (a, d) = 1$ otherwise $k \mid a$, $k \mid a+b$

  therefore $k \mid b$

- then by Euclid Lemma $d \mid p$

  $d$ is either $1$ or $p$

III. numbers of primes are infinite.

- first $r$ primes $p_1, p_2 \cdots p_r$
- $p_1 p_2 \cdots p_r + 1$ is not divisible by $p_1$ or $p_2$ or $\cdots$ $p_r$
- for any finite $r$, the number of prime is larger than $r$

---

IV.

**16.** Find an integer $n$ such that $n/2$ is a square, $n/3$ is a cube, and $n/5$ is a fifth power.

suppose $n = 2^x 3^y 5^z$ then

$x - 1 \equiv y \equiv z \equiv 0 \pmod 2$

$x \equiv y - 1 \equiv z \equiv 0 \pmod 3$

$x \equiv y \equiv z - 1 \equiv 0 \pmod 5$

apply CRT

$x \equiv 15 \pmod{30}$

$y \equiv 10 \pmod{30}$

$z \equiv 6 \pmod{30}$

unique solution

if primes other than 2, 3, 5 cannot divide $n$.

$n = 2^{15} \cdot 3^{10} \cdot 5^6$

***41.** Prove that an odd integer $n > 1$ is a prime if and only if it is not expressible as a sum of three or more consecutive positive integers.

- (n) $= (2m+d-1)\cdot d/2 \quad (m \geq 1,\ d \geq 3)$

---

- Claim: $n > 1$ odd 일 때

  $n$ is not prime $\iff 2n = (2m+d-1)\cdot d \quad (m \geq 1, d \geq 3)$

---

- Claim: $n = (2m+d-1)\cdot d \iff n = ab$ ① $a, b$ 홀짝성 다름.
  ② $3 \leq \min(a,b)$

  $(m \geq 1,\ d \geq 3)$

($\Rightarrow$) ① $d$와 $(2m+d+1)$는 홀짝성 다름.

② $3 \leq d \leq 2m+d-1$

($\Leftarrow$) $\min(a,b) = d$

$\max(a,b) = d + (2m-1)$

$= d+1, d+3, d+5 \ldots$

홀짝성이 다른 모든 경우를 포함.

---

- Claim: $n > 1$ odd 일 때

  $n$ is not prime $\iff 2n = ab$ ① $a, b$ 홀짝성 다름.
  ② $3 \leq \min(a,b)$

case ① $n$ is prime

(모순증명) $2n = 2\cdot(n)$ (prime)

i.e $2n$을 ①, ② 조건을 만족하게 쪼갤 수 없음.

case ② $n$ is composite.

$n$을 나누는 가장 작은 $\Rightarrow$

홀수 prime $p$

① $p$와 $2n/p$ 홀짝성 다름.

② $3 \leq p \leq \frac{2n}{p}$ → $p$ 이상의 다른 홀수 포함

인수로 포함됨.

**42.** If $2^n + 1$ is an odd prime for some integer $n$, prove that $n$ is a power of 2. (H)

- if $n$ is not power of 2.

  $n = ab$ with ① $a, b \in \mathbb{Z}$

  ② $b > 1$ odd

  ③ $a \geq 1$

- $(2^a)^b + 1 = ((2^a) + 1)((2^a)^{b-1} - (2^a)^{b-2} \cdots + 1)$

- since $2^a + 1 < 2^{ab} + 1$ because $a < ab$ (∵ ②)

  $((2^a)^{b-1} - (2^a)^{b-2} \cdots + 1) \neq 1$ and $2^4 + 1$ is not prime

---

**43.** The numbers $F_n = 2^{2^n} + 1$ in the preceding problem are called the *Fermat numbers*, after Pierre Fermat who thought they might all be primes. Show that $F_5$ is composite by verifying that

$$(2^9 + 2^7 + 1)(2^{23} - 2^{21} + 2^{19} - 2^{17} + 2^{14} - 2^9 - 2^7 + 1) = 2^{32} + 1.$$

(It is not hard to show that $F_n$ is prime for $n = 0, 1, \cdots, 4$; these are the only $n$ for which $F_n$ is known to be prime. It is now known that $F_n$ is composite for $n = 5, 6, \cdots, 21$. It is conjectured that only finitely many Fermat numbers are prime.)

$$2^{32} - 2^{30} + 2^{28} - 2^{26} + 2^{23} - 2^{18} - 2^{16} + 2^9$$
$$+ 2^{30} - 2^{28} + 2^{26} - 2^{24} + 2^{21} - 2^{16} - 2^{14} + 2^7$$
$$+ 2^{23} - 2^{21} + 2^{19} - 2^{17} + 2^{14} - 2^9 - 2^7 + 1$$

**4.** If $2^n - 1$ is a prime for some integer $n$, prove that $n$ is itself a prime. (Numbers of the form $2^p - 1$, where $p$ is a prime, are called the *Mersenne numbers* $M_p$ because the Frenchman Father Marin Mersenne (1588–1648) stated the $M_p$ is a prime for $p = 2, 3, 5, 7, 13, 17, 19, 31, 67, 127, 257$, but is composite for all other primes $p < 257$. It took some 300 years before the details of this assertion could be checked completely, with the following outcome: $M_p$ *is not* a prime for $p = 67$ and $p = 257$, and $M_p$ *is* a prime for $p = 61$, $p = 89$, and $p = 107$. Thus, there are 12 primes $p < 257$ such that $M_p$ is a prime. It is now known that $M_p$ is a prime in the following additional cases, $p = 521, 607, 1279, 2203, 2281, 3217, 4253, 4423, 9689, 9941, 11213, 19937, 21701, 23209, 44497, 86243, 110503, 132049, 216091$. The Mersenne prime $M_{216091}$ is the largest specific number that is known to be prime. It is conjectured that infinitely many of the Mersenne numbers are prime.)

- $n = 1$ then $2^n - 1 = 1$
- $n$ is composite then $n = ab$ $(a \geq 2,\ b \geq 2)$

$$(2^a)^b - 1 = (2^a - 1)\left((2^a)^{b-1} + (2^a)^{b-2} + \cdots + 1\right)$$

- $3 \leq 2^n - 1 < 2^{ab} - 1$ (because $2 \leq a < ab$)
- $2^{ab} - 1$ divisible by $(2^n - 1)$

therefore not prime

***48.** Prove that there are infinitely many primes by considering the sequence $2^{2^1}+1, 2^{2^2}+1, 2^{2^3}+1, 2^{2^4}+1, \cdots$. (H)

$A_1$ $A_2$ $A_3$ . . .

---

- $2^{2^m}-1 = A_m - 2 = 3 \cdot (2^{2^1}+1) \cdots (2^{2^{m-1}}+1)$ $(m \geq 1)$

$A_1$ $\quad\quad$ $A_{m-1}$

---

Claim $(A_n, A_m) = 1$ $(n \neq m)$

- otherwise $d \mid A_n$, $d \mid A_m$ $(n<m)$ $(d>1)$

- $A_m - 2 = 3 A_1 \cdots A_n \cdots A_{m-1}$

- $d$ divides both $A_m$, $A_m - 2$

  i.e $d \leq 2 \Rightarrow d = 2$

- but $A_n = 2^{2^n}+1$ is odd. $\longrightarrow\!\longleftarrow$ ☹.

---

$A_1, A_2, \ldots, A_n$ are pairwise coprime.

each $A_n$ implies new prime

**9.** Let $f(x)$ be a function of a real variable, and let $\Delta f$ be the function $\Delta f(x) = f(x+1) - f(x)$. For $k > 1$, put $\Delta^k f = \Delta(\Delta^{k-1} f)$. The function $\Delta^k f(x)$ is called the $k$th *forward difference* of $f$. Show that

$$\Delta^k f(x) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} f(x+k-j).$$

- $k=1$ $\quad \Delta f(x) = f(x+1) - f(x)$

  $j=0 \quad j=1$

- suppose equality holds at $k=n$

  ① $\Delta^{n+1} f(x) = \Delta^n f(x+1) - \Delta^n f(x)$

  ② $\Delta^n f(x) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} f(x+n-j)$

then $\Delta^{n+1} f(x) = \Delta^n f(x+1) - \Delta^n f(x)$

$$= \sum_{j=0}^{n} (-1)^j \binom{n}{j} \{ f(x+1+n-j) - f(x+n-j) \}$$

$n+1 \cdots 1 \qquad n \cdots 0$

$$= \sum_{j=-1}^{n-1} (-1)^{j+1} \binom{n}{j+1} f(x+n-j) - \sum_{j=0}^{n} (-1)^j \binom{n}{j} f(x+n-j)$$

because $\binom{n}{j+1} + \binom{n}{j} = \binom{n+1}{j+1}$

$$= \sum_{j=0}^{n-1} (-1)^{j+1} \binom{n+1}{j+1} f(x+n-j)$$

$$+ f(x+n+1) + (-1)^{n+1} f(x)$$

$j=-1 \qquad j=n$

let $k = j+1$ then

$$= \sum_{k=0}^{n+1} (-1)^k \binom{n+1}{k} f(x+n+1-k)$$

***11.** Show that $\binom{x}{k}$ is a polynomial in $x$ of degree $k$ and leading coefficient $1/k!$. Let $P(x)$ be an arbitrary polynomial with real coefficients and degree at most $n$. Show that there exist real numbers $c_k$ such that

$$P(x) = \sum_{k=0}^{n} c_k \binom{x}{k} \tag{1.16}$$

for all $x$, and that such $c_k$ are uniquely determined.

if $n=0$ 일때 $P(x) = a$. let $c_0 = a$

if 가정이 $n-1$까지에 성립.

$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \quad (a_n \neq 0)$

$Q(x) = P(x) - a_n \cdot n! \binom{x}{n}$ 는 $n-1$차이므로 $\sum_{k=0}^{n-1} c_k \binom{x}{k}$

형태로 유일하게 나타난다.

$\Rightarrow P(x) = Q(x) + a_n \cdot n! \binom{x}{n}$ 이고,

각 차수별로 계수를 일치하는 것이 유일하다.

***16.** Show that if $f(x)$ is a polynomial of degree $n$ with real coefficients, which takes integral values on a certain set of $n+1$ consecutive integers, then $f(x)$ is integer-valued.

claim · suppose $g(x)$ takes int values

on $x = 0, 1, \ldots, n$ (degree $n$)

then $g(x)$ is int-valued

- $g(x) = \sum_{k=0}^{n} c_k \binom{x}{k}$

⓪ $g(0) = c_0$ (정수 = 정수)

① $g(1) = c_0 + c_1$ (정수 = 정수 + 정수)

⋮

ⓙ $g(j) = \sum_{k=0}^{j-1} c_k \binom{j}{k} + c_j \binom{j}{j}$ (정수 = 정수 (귀납가정) + 따라서 정수)

$\Rightarrow$ $c_0, c_1, \ldots, c_n$은 정수.

$\Rightarrow$ $\forall x \in \mathbb{Z}$ then $f(x) \in \mathbb{Z}$

있으니 $f(x)$를 귀납가정이용하면 $g(x)$의 식으로 만들 수 있음.

UI.

**2.** Exhibit a complete residue system modulo 17 composed entirely of multiples of 3.

$S = \{0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48\}$

**6.** Prove that if $p$ is a prime and $a^2 \equiv b^2 \pmod{p}$, then $p|(a+b)$ or $p|(a-b)$.

$p \mid a^2 - b^2 = (a+b)(a-b)$

if $p \nmid (a+b)$ then by Euclid Lemma $p \mid a-b$

i.e $(p, a+b) = 1$

**8.** Prove that any number that is a square must have one of the following for its units digit: 0, 1, 4, 5, 6, 9.

$\forall n \in \mathbb{N}$ ($n^2$의 일의자리 수) $\equiv n^2 \pmod{10}$

$0^2 \equiv 0$ $1^2 \equiv 1$ $2^2 \equiv 4$ $3^2 \equiv 9$ $4^2 \equiv 6$

$5^2 \equiv 5$ $6^2 \equiv 6$ $7^2 \equiv 9$ $8^2 \equiv 4$ $9^2 \equiv 1 \pmod{10}$

**14.** Show that $7|(3^{2n+1} + 2^{n+2})$ for all $n$.

$$3^{2n+1} + 2^{n+2} \equiv 3 \cdot 9^n + 4 \cdot 2^n$$

$$\equiv 3 \cdot 2^n + 4 \cdot 2^n$$

$$\equiv 7 \cdot 2^n$$

$$\equiv 0 \pmod{7}$$

**20.** Prove that $n^7 - n$ is divisible by 42, for any integer $n$.

- $7 \mid n^7 - n$ for $\forall n \in \mathbb{N}$,

  because $n^7 \equiv n \pmod 7$ (Fermat)

- $3 \mid n^7 - n$ for $\forall n \in \mathbb{N}$

  ① $n^3 \equiv n \pmod 3$ (Fermat)

  ② $n^7 - n \equiv n^3 - n$

  $\equiv n - n$

  $\equiv 0 \pmod 3$

- $2 \mid n^7 - n$ for $\forall n \in \mathbb{N}$

  ① $n^2 \equiv n \pmod 2$

  ② $n^7 - n \equiv n^4 - n$

  $\equiv n^2 - n$

  $\equiv n - n \equiv 0 \pmod 2$

---

**27.** Prove that $\frac{1}{5}n^5 + \frac{1}{3}n^3 + \frac{7}{15}n$ is an integer for every integer $n$.

$$\frac{1}{5}(n^5 - n) + \frac{1}{3}(n^3 - n) + n$$

since $n^5 \equiv n \pmod 5$ $\quad n^3 \equiv n \pmod 3$

$5 \mid n^5 - n$ $\quad 3 \mid n^3 - n$

$\frac{n^5 - n}{5}, \frac{n^3 - n}{3} \in \mathbb{Z}$

**36.** If $p$ is a prime, prove that $(p-1)!+1$ is a power of $p$ if and only if $p = 2, 3$, or $5$. (H)

$p=2$ then $(2-1)!+1=2$

$p=3$ then $(3-1)!+1=3$

$p=5$ then $(5-1)!+1=25$.

otherwise if $p \geq 7$ then $p^k = (p-1)!+1$

$p^k - 1 = (p-1)(p^{k-1} + p^{k-2} + \cdots + 1) = (p-1)!$

then $p^{k-1} + \cdots + 1 = (p-2)!$

LHS $\equiv k \pmod{p-1}$ because $p \equiv 1 \pmod{p-1}$

RHS $\equiv (p-2)!$ because $p-1$ is composite.

$\equiv 0 \pmod{p-1}$

therefore $p-1 \leq k$ and $(p-1)! + 1 < p^{p-1} \leq p^k$.

***45.** Show that if $p$ is prime then $\binom{p}{k} \equiv 0 \pmod{p}$ for $1 \leq k \leq p-1$.

- $\dfrac{p!}{k!(p-k)!}$ then
  - ($\geq 2$) $p \mid p!$
  - ($\leq 2$) $p \nmid (k)!$ and $p \nmid (p-k)!$

    because $k < p$ $p-k < p$

- meanwhile $\dfrac{p!}{k!(p-k)!} \in \mathbb{Z}$ therefore $p \mid \binom{p}{k}$

  $\binom{p}{k} \equiv 0 \pmod{p}$

Note. $(x+y)^p \equiv x^p + \binom{p}{1} x^{p-1}y + \cdots + \binom{p}{p-1} xy^{p-1} + y^p$

$\equiv x^p + y^p \pmod{p}$